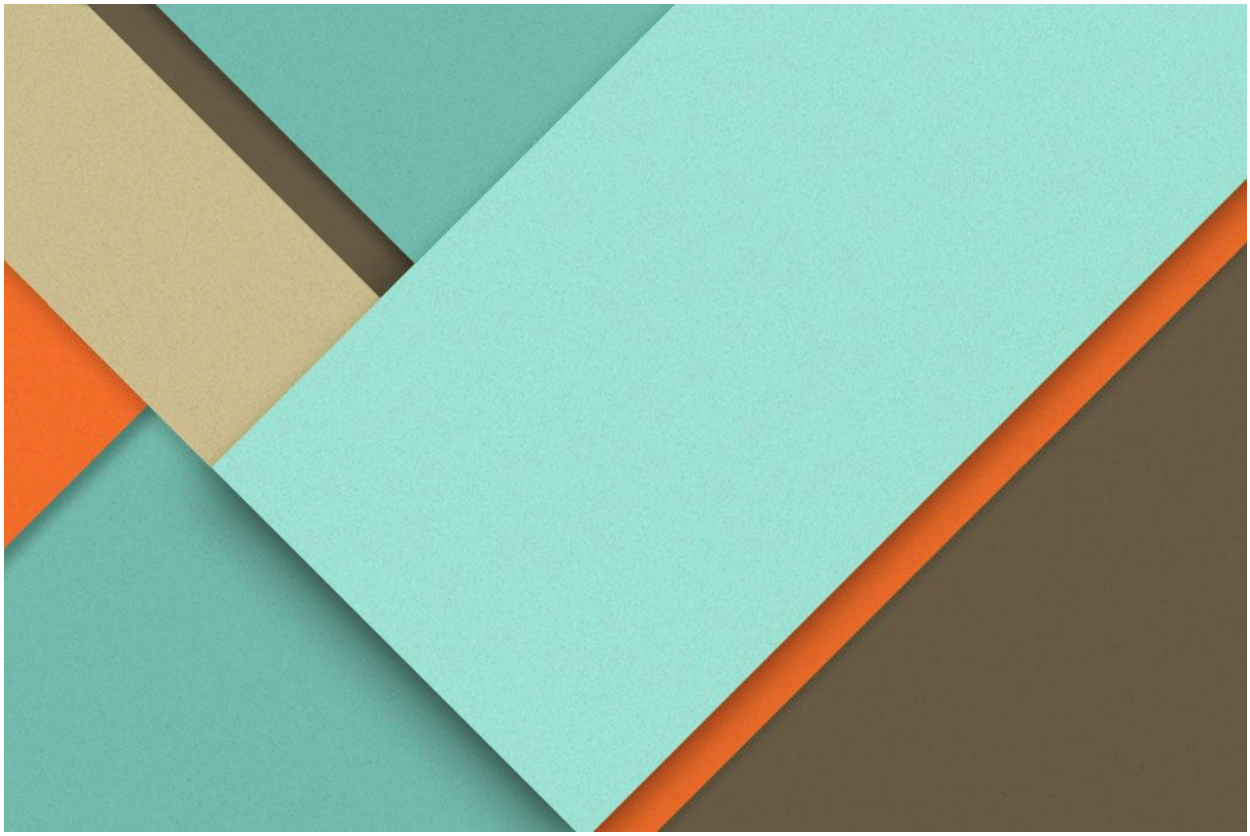

Loan Eligibility Prediction

—

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Springboard

Overview

In today's rapidly evolving financial landscape, the demand for loans has increased significantly as individuals and businesses rely on credit to meet personal, educational, housing, and entrepreneurial needs. Financial institutions receive a large volume of loan applications daily, making it challenging to evaluate each applicant quickly while maintaining accuracy and fairness. As a result, there is a growing need for intelligent tools that can assist in the preliminary screening of loan applicants.

The Quick Loan Eligibility Predictor is designed to address this challenge by providing a fast and data-driven method to estimate whether a loan applicant is likely to qualify for a loan. Using historical financial data and key applicant attributes such as income, employment status, credit history, and property area, the system can generate a prediction that helps financial institutions quickly assess the potential eligibility of an applicant.

Goal and Audience

The goal of this project is to build a predictor that can classify loan approval status (Loan_Status). Financial services companies could benefit from utilizing this predictor to quickly screen applications and minimize applications that require in depth manual review and approval.

Data Source

Existing data was sourced from Kaggle with the following data points:

Column	Description	Example
Customer_ID	Unique identifier for each loan applicant	569
Gender	Gender of the applicant	Male / Female

Married	Marital status of the applicant	Yes / No
Dependents	Number of dependents	0, 1, 2, 3
Education	Education level of the applicant	Graduate / Not Graduate
Self_Employed	Whether the applicant is self-employed	Yes / No
Applicant_Income	Applicant's monthly income	5000
Coapplicant_Income	Coapplicant's monthly income	1500
Loan_Amount	Loan amount requested (in thousands)	128
Loan_Amount_Term	Loan repayment term (in months)	360
Credit_History	Credit history meets lending criteria (1 = Yes, 0 = No)	1
Property_Area	Type of property area	Urban / Semiurban / Rural
Loan_Status	Loan approved or not (target variable)	Y / N

The Loan_Status column is the dependent variable that we are trying to predict.

Data Wrangling

I. Feature Engineering

The original data set was in a great condition and did not require a lot of cleaning.

There are some features that could be added to aid with data analysis and potentially modelling our predictor. These are the added features below that were missing from original data set that are evidently used by the financial sector during loan application screening:

- Total_Income as the sum of Applicant_Income and Coapplicant_Income
- Monthly_Payment = Full_Loan_Amount (in thousands) / Loan_Amount_Term
- Debt_To_Income_Ratio as a percentage of Monthly_Payment / Total_Income

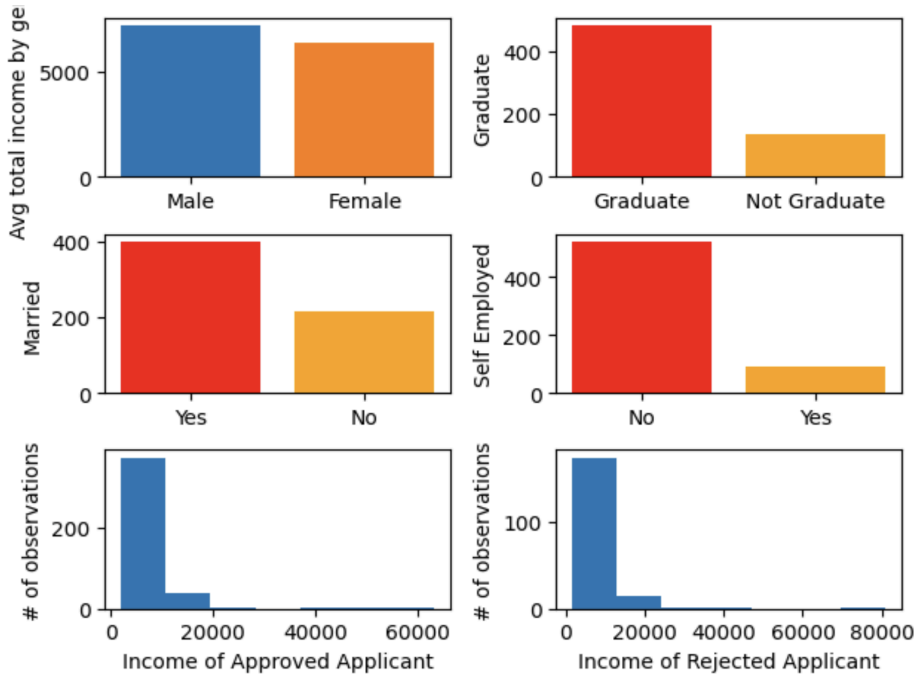
II. Data Cleaning and Processing

The following data processing was done for the data set:

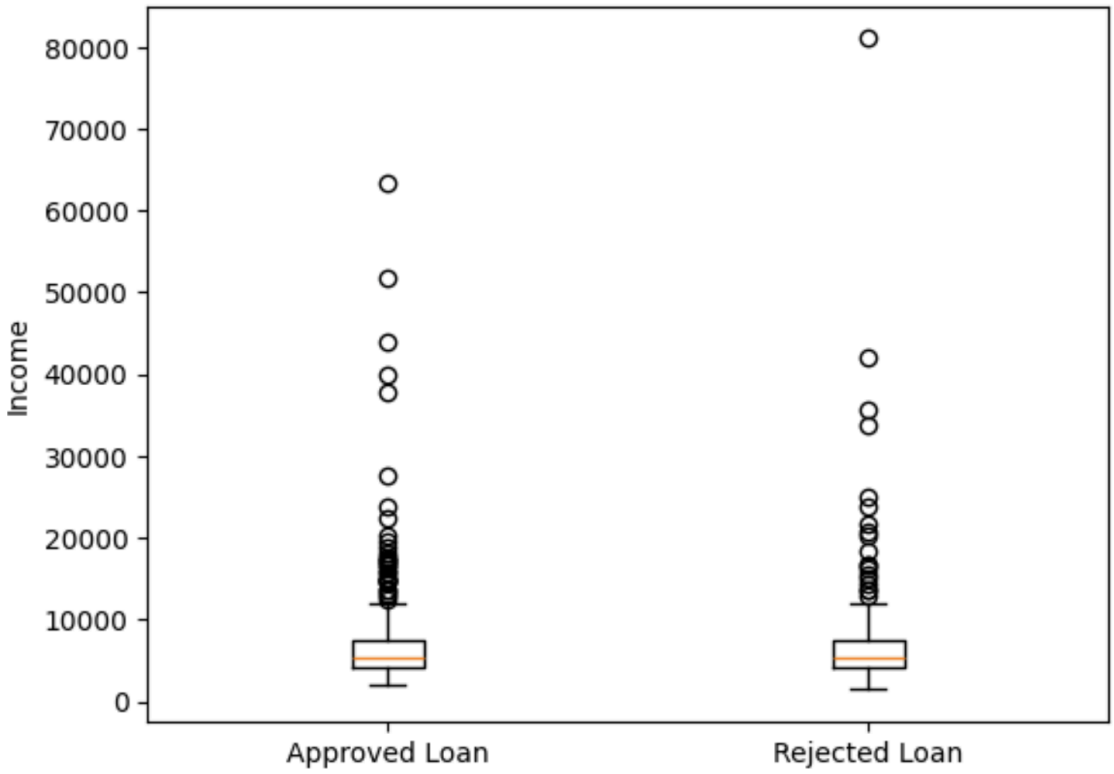
- One hot encoding for categorical features such as Property_Area
- Mapped binary categorical variables to numeric 0s and 1s such as Gender, Education Status, Marital Status, Loan Status, etc.
- Scaled the numeric columns for modelling
- Removed two records that were deemed bad data (more in data analysis below)

III. Data Analysis

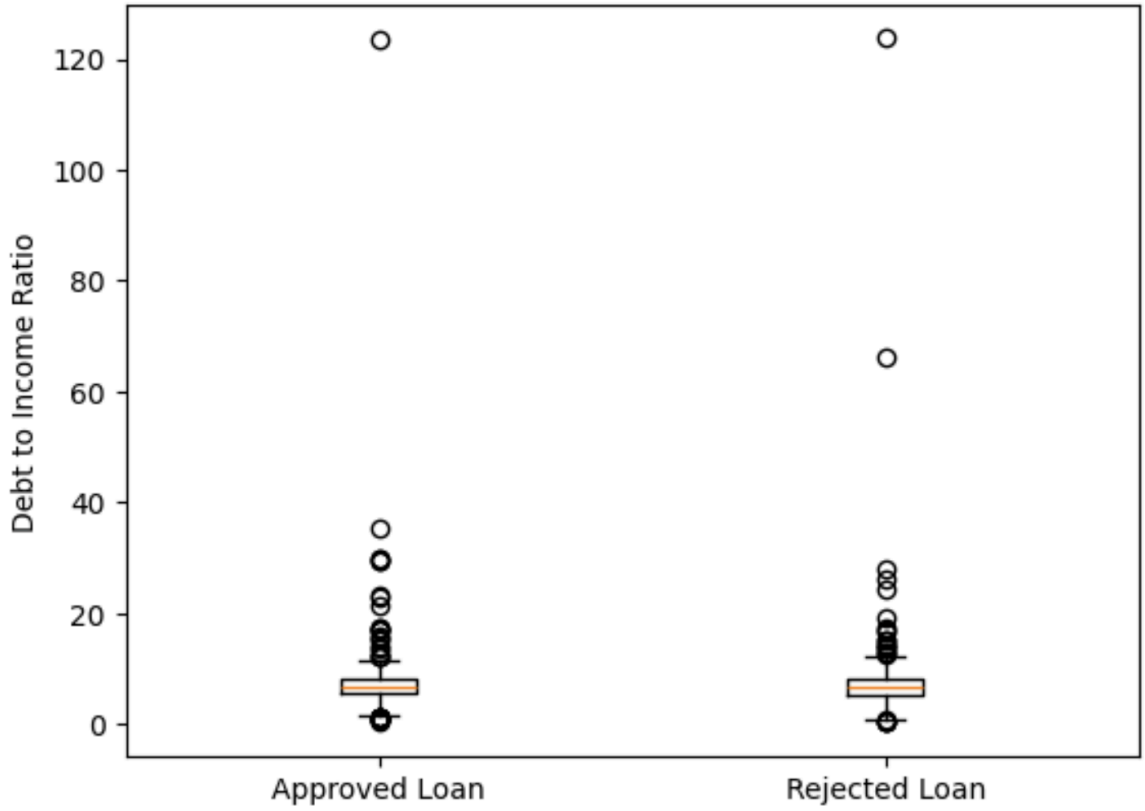
Initial investigation started with slicing the data to see the distribution between features:



Initial investigation also shows income alone does not dictate loan approval which is a fact and the reason why we added a Debt_To_Income_Ratio data point:



With the assumption that Debt_To_Income_Ratio and Credit_History columns are two of the most important consideration dictation approval status, a quick charting of its values sliced by approval status revealed some outliers:



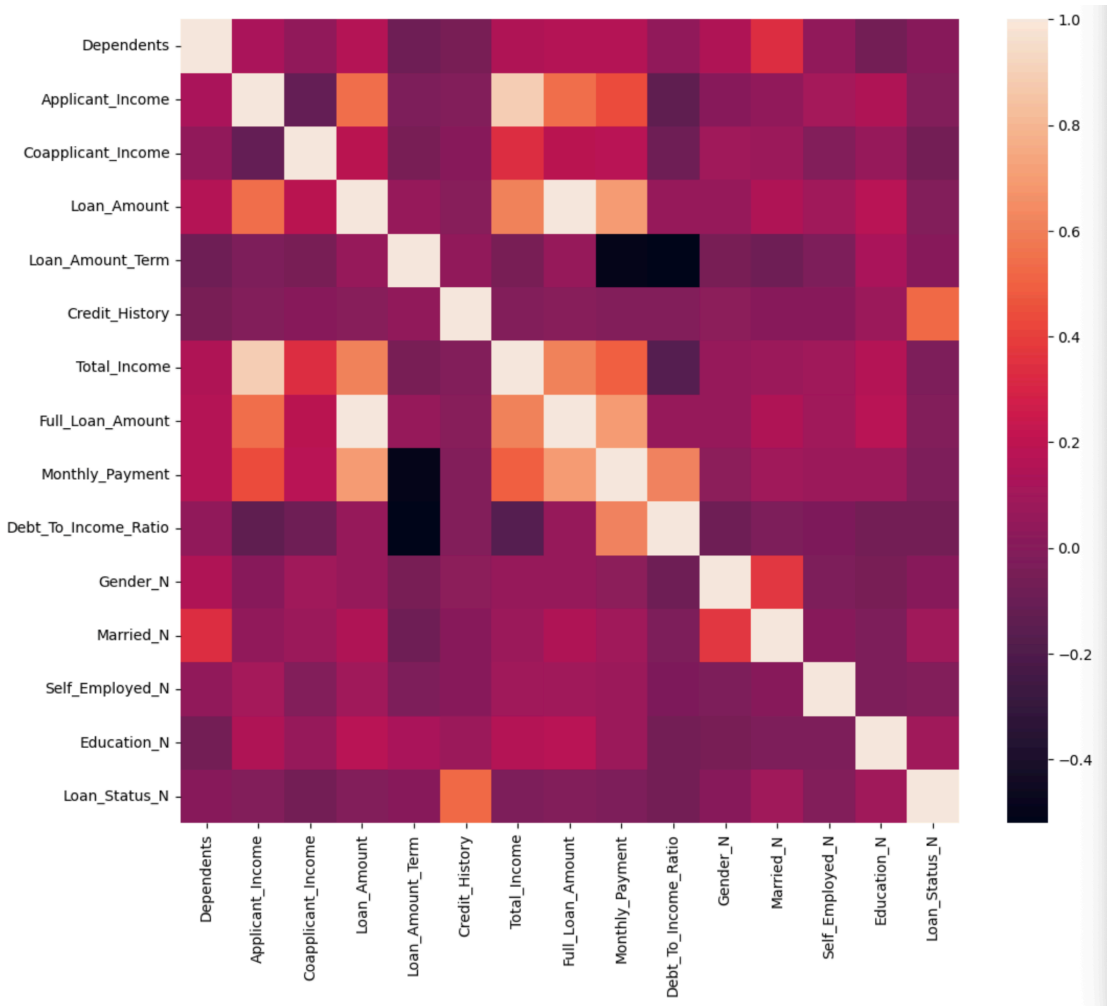
A loan approved record with id: 498 has a debt to income ratio that is over 100% which is odd as one can assume a loan with a monthly payment over the declared monthly income could not have been approved. In addition this record also has no credit history.

Before deciding to drop this record, a verification was done on the debt_to_income ratio of other approved records with no credit history which revealed that approved loans without a credit history only have a Debt_To_Income_Ratio < 11%. This record was then dropped from the data set.

Customer_ID	Total_Income	Full_Loan_Amount	Loan_Amount_Term	Monthly_Payment	Debt_To_Income_Ratio	Loan_Status	Credit_History
498	7482.0	111000	12	9250.00	123.63	Y	0
454	4949.0	90000	180	500.00	10.10	Y	0
202	4923.0	166000	360	461.11	9.37	Y	0
319	3541.0	112000	360	311.11	8.79	Y	0
156	39999.0	600000	180	3333.33	8.33	Y	0
327	4917.0	130000	360	361.11	7.34	Y	0
268	6144.0	160000	360	444.44	7.23	Y	0
528	6715.0	161000	360	447.22	6.66	Y	0
199	6232.0	113000	360	313.89	5.04	Y	0
123	11117.0	137000	360	380.56	3.42	Y	0

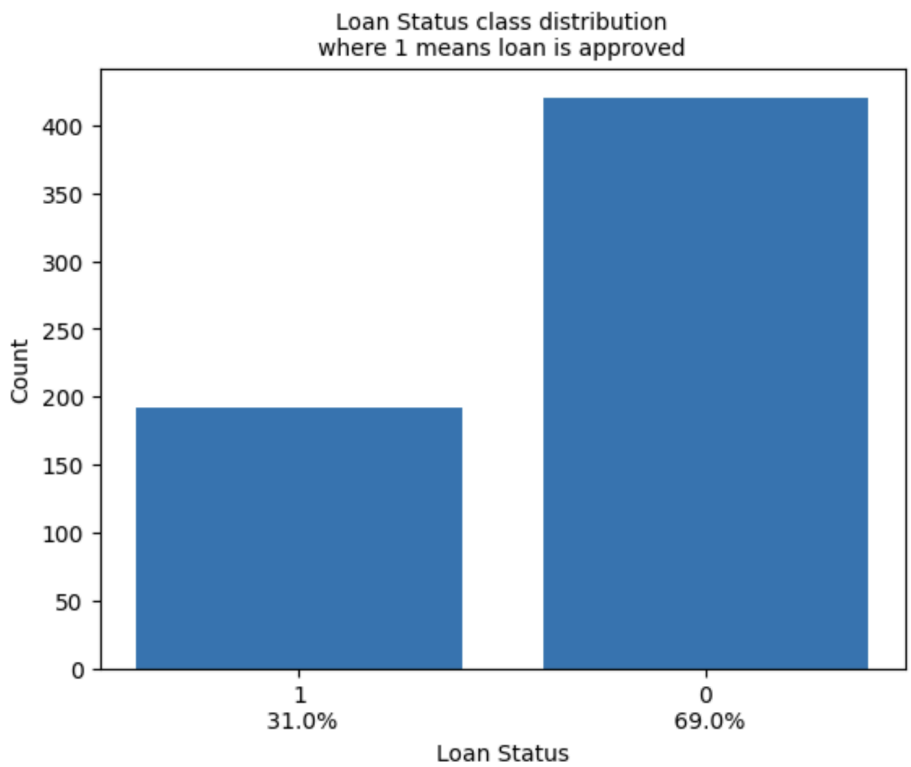
Figure above shows approved records with no credit history sorted by Debt_To_Income_Ratio descending.

A correlation matrix revealed Credit_History is strongly correlated with Loan_Status:



Modelling

The original data set is skewed with more data representing loan rejection status as shown below. This required stratifying our dependent variable when splitting the training and test sets.



I. Logistic Regression

GridSearchCV was utilized for best parameter discover which revealed the following:
Best parameters: {'C': 10, 'penalty': 'l1', 'solver': 'liblinear'}

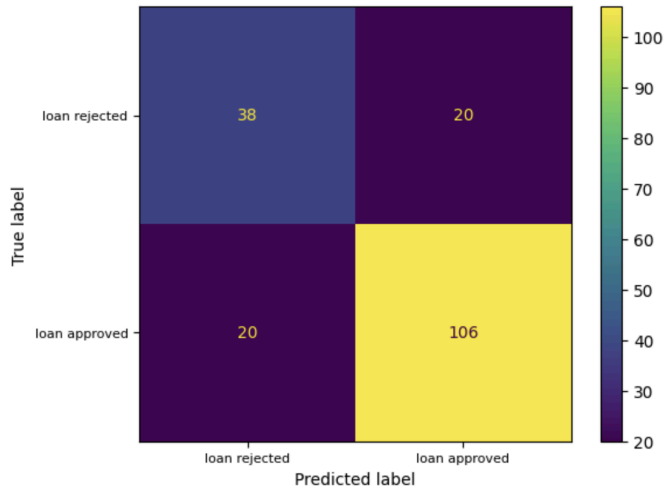
Accuracy test on the test data = 0.78

```
Logistic Regression: Classification Report for Test Data
              precision    recall  f1-score   support

           0       0.66      0.66      0.66        58
           1       0.84      0.84      0.84       126

    accuracy                           0.78       184
   macro avg       0.75      0.75      0.75       184
weighted avg       0.78      0.78      0.78       184
```

Confusion matrix:



II. Random Forest

Random Forest: Accuracy=0.826

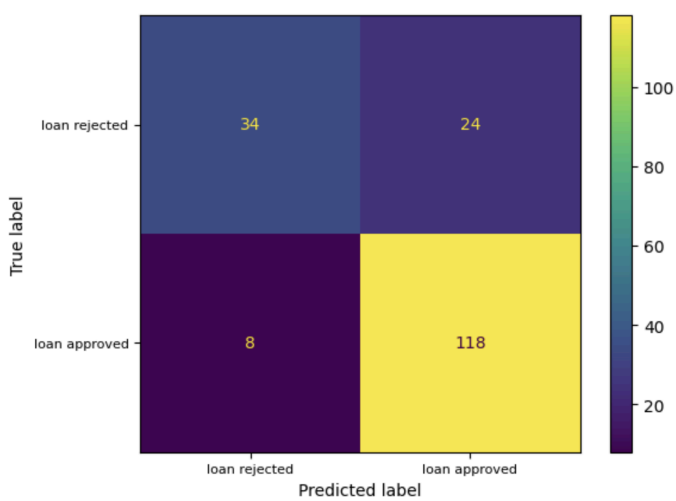
Random Forest: f1-score=0.817

```
Random Forest: Classification Report for Test Data
              precision    recall  f1-score   support

           0       0.81      0.59      0.68        58
           1       0.83      0.94      0.88       126

    accuracy                           0.83       184
   macro avg       0.82      0.76      0.78       184
weighted avg       0.82      0.83      0.82       184
```

Confusion matrix:

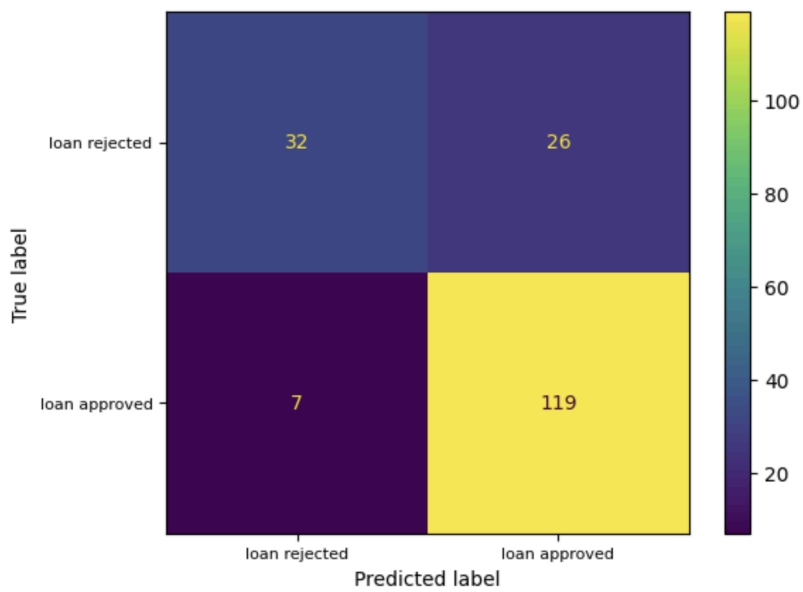


III. Gradient Boosting

Accuracy: 0.82

```
Gradient Boosting: Classification Report for Test Data
              precision    recall  f1-score   support

           0       0.82      0.55      0.66        58
           1       0.82      0.94      0.88       126

    accuracy                           0.82       184
   macro avg       0.82      0.75      0.77       184
weighted avg       0.82      0.82      0.81       184
```

Confusion matrix: